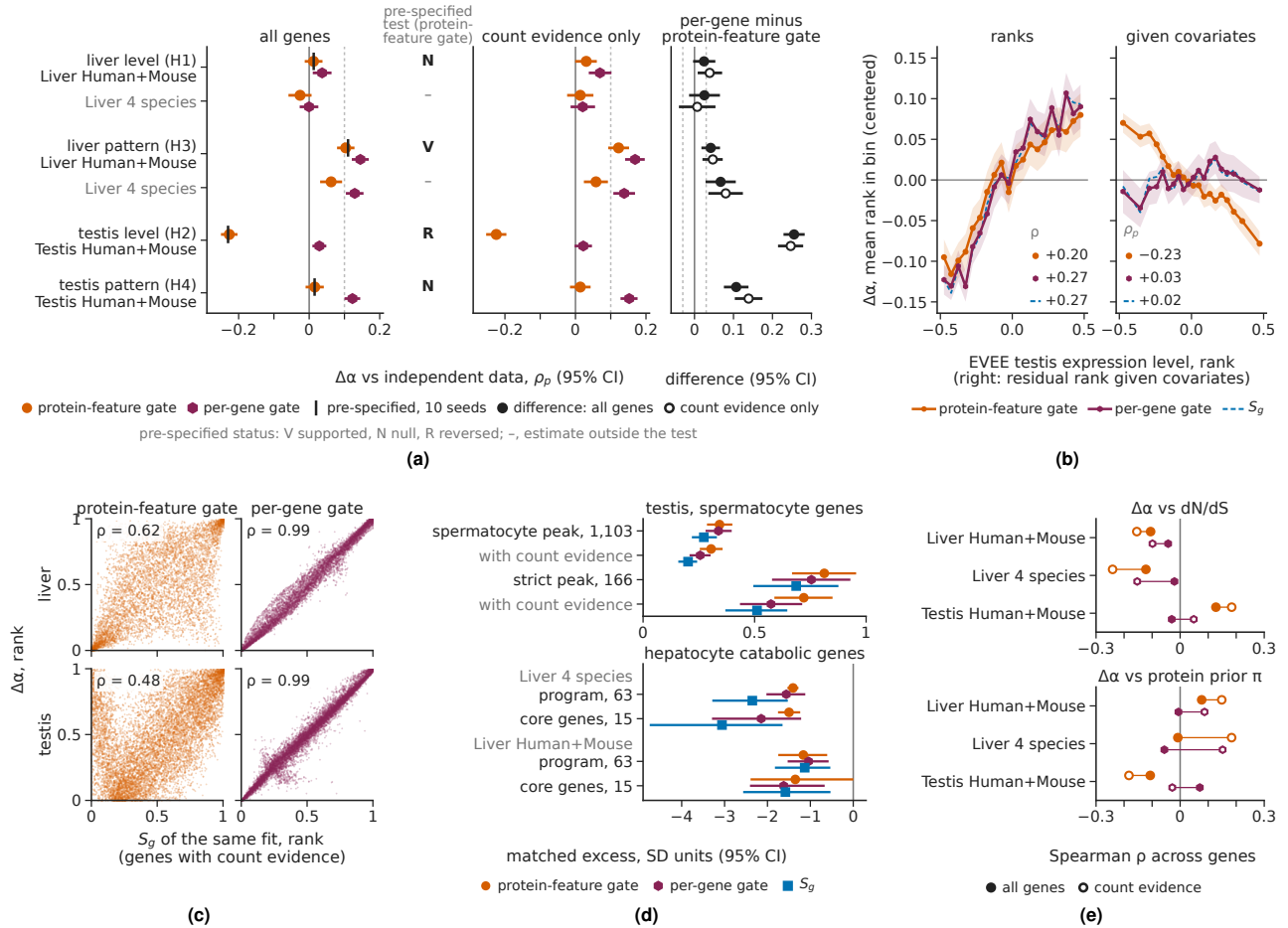

all genes
pre-specified test (protein-feature gate)
count evidence only
per-gene minus protein-feature gate
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
testis pattern (H4)
Testis Human+Mouse
N
–
V
–
R
N
−0.2
0
0.2
0
0.1
0.2
0.3
Δα vs independent data, $\rho_p$ (95% CI)
difference (95% CI)
protein-feature gate
per-gene gate
pre-specified, 10 seeds
difference: all genes
count evidence only
pre-specified status: V supported, N null, R reversed; –, estimate outside the test
(a)
ranks
given covariates
Δα, mean rank in bin (centered)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.5
0.0
0.5
ρ
+0.20
+0.27
+0.27
$\rho_p$
−0.23
+0.03
+0.02
EVEE testis expression level, rank
(right: residual rank given covariates)
protein-feature gate
per-gene gate
$S_g$
(b)
protein-feature gate
per-gene gate
liver
testis
Δα, rank
ρ = 0.62
ρ = 0.99
ρ = 0.48
ρ = 0.99
0
0.5
1
$S_g$ of the same fit, rank
(genes with count evidence)
(c)
testis, spermatocyte genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
hepatocyte catabolic genes
Liver 4 species
program, 63
core genes, 15
Liver Human+Mouse
program, 63
core genes, 15
−4
−3
−2
−1
0
matched excess, SD units (95% CI)
protein-feature gate
per-gene gate
$S_g$
(d)
Δα vs dN/dS
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
−0.3
0
0.3
Δα vs protein prior π
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
Spearman ρ across genes
all genes
count evidence
(e)